

Experiment No: 01

Experiment Name: Study the Role of Power Electronics for Environment and Sustainability

Abstract

Power electronics — the branch of electrical engineering concerned with the conversion and control of electric power using semiconductor switching devices such as diodes, thyristors, power MOSFETs, and IGBTs — has become a cornerstone technology for achieving global energy and environmental sustainability. This report studies the role that power-electronic converters play in enabling renewable-energy integration, smart-grid operation, electric transportation, and industrial energy efficiency, thereby reducing greenhouse-gas emissions and supporting a cleaner energy future. Recent advances in wide-bandgap semiconductors, particularly silicon carbide (SiC) and gallium nitride (GaN), have pushed converter efficiencies beyond 98 percent, and the global power-electronics market for renewable-energy applications is projected to grow from roughly USD 16-18 billion in 2025 to more than USD 40 billion by the mid-2030s.

The discussion combines general concepts with country-specific data for Bangladesh, drawing on figures published by the International Energy Agency (IEA), the Bangladesh Power Development Board (BPDB), the Sustainable and Renewable Energy Development Authority (SREDA), and Power Grid Company of Bangladesh PLC (PGCB), to illustrate the present state of the power sector, the extent of renewable-energy penetration, and the challenges of meeting rising demand while the national grid still depends heavily on natural gas, oil, and coal.

Introduction

Power electronics is the application of electronics to the control and conversion of electric power. Early high-power conversion relied on mercury-arc valves; modern systems instead use semiconductor switching devices, with variable-speed drive systems ranging in power from a few hundred watts to tens of megawatts. Because almost every stage of modern power generation, transmission, and consumption now passes through some form of power-electronic converter, the technology sits at the centre of the transition toward a more sustainable energy system.

Climate change and the continuous growth of global electricity demand have made efficient energy generation, transmission, and utilisation one of the greatest engineering challenges of this century. Beyond traditional motor drives and uninterruptible power supplies, power electronics is now essential for integrating renewable sources, electric vehicles, HVDC transmission, and smart grids, all of which depend on converters for voltage regulation, frequency control, and grid synchronisation.

Energy sustainability refers to meeting present and future energy demand without exhausting natural resources, primarily through sources such as solar, wind, geothermal, hydropower, and ocean energy that are generally available at no fuel cost and can be used repeatedly. Environmental sustainability, as defined by the United Nations World Commission on Environment and Development, means acting in a way that ensures future generations retain

access to the natural resources needed for a quality of life at least equal to that of the present generation, while improving human welfare within the carrying capacity of the Earth's supporting ecosystems.

Key Features of Power Electronics for Sustainability

Power-electronic converters act as the interface that transfers electricity from one stage of the supply chain to the next while allowing its voltage, frequency, and waveform to be flexibly controlled. This flexibility underlies several features that make power electronics central to environmental and energy sustainability:

- **Renewable-energy integration:** Inverters convert the DC output of solar photovoltaic panels and battery systems, and back-to-back converters regulate variable-frequency wind power, into stable grid-compatible AC, allowing intermittent sources to be connected safely to the grid.
- **Wide-bandgap devices:** Silicon carbide (SiC) and gallium nitride (GaN) semiconductors switch faster and operate at higher voltage and temperature than conventional silicon, cutting conversion losses and reaching efficiencies close to 99 percent.
- **Smart-grid and HVDC integration:** Power-electronic interfaces allow bidirectional power flow, voltage regulation, and grid-stability support as more variable renewable capacity is added, giving grid operators far more flexible control over voltage, frequency, and power flow.
- **Electrification of transport:** Traction inverters, on-board chargers, battery-management systems, and DC-DC converters manage power flow between batteries, motors, and charging systems in electric, hybrid, and increasingly electric-aircraft applications.
- **Green industry and motor drives:** Variable-frequency drives (VFDs) and high-efficiency converters minimise conversion losses and cut unnecessary energy consumption in industrial motors, pumps, and compressors.
- **Energy storage and microgrids:** Bidirectional converters regulate battery charging and discharging, supporting grid stability, DC microgrids for rural electrification, and islanded microgrid operation during disturbances.

Data

This section presents quantitative data on the global power-electronics market and converter efficiency, together with the current operating condition of Bangladesh's national grid. Global market and technology figures are drawn from industry reports; Bangladesh country-level figures are compiled from IEA data services, while real-time grid figures were obtained from BPDB and the PGCB National Load Dispatch Centre (NLDC) ERP portal.

Global Market and Technology Trends

The global power-electronics market for renewable-energy applications was valued at approximately USD 16-18 billion in 2025 and is projected to grow at 8-9 percent CAGR, exceeding USD 40 billion by the mid-2030s, reflecting the rapid expansion of solar, wind, and storage installations worldwide.

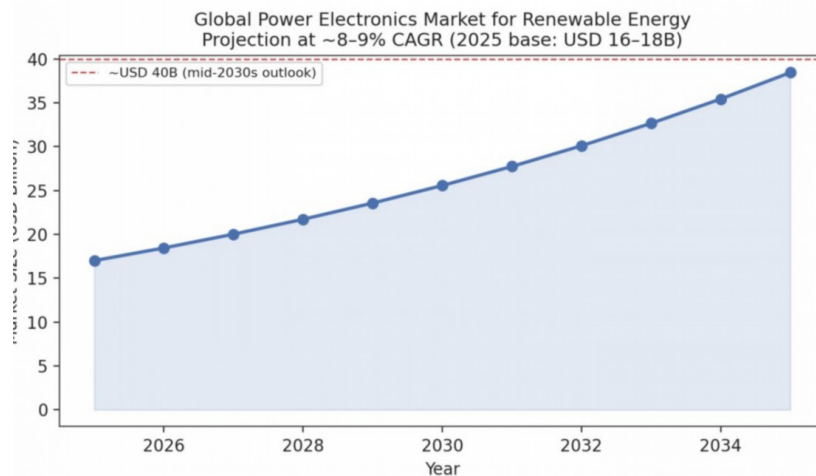


Figure 1. Projected growth of the global power-electronics market for renewable-energy applications (~8-9% CAGR from a 2025 base of USD 16-18 billion).

Wide-bandgap converters based on SiC and GaN already reach 98.8-98.9 percent conversion efficiency, compared with 95-96 percent for conventional silicon devices, as shown in Figure 2. This efficiency gain directly reduces heat losses and energy waste at every conversion stage, from solar inverters to EV chargers.

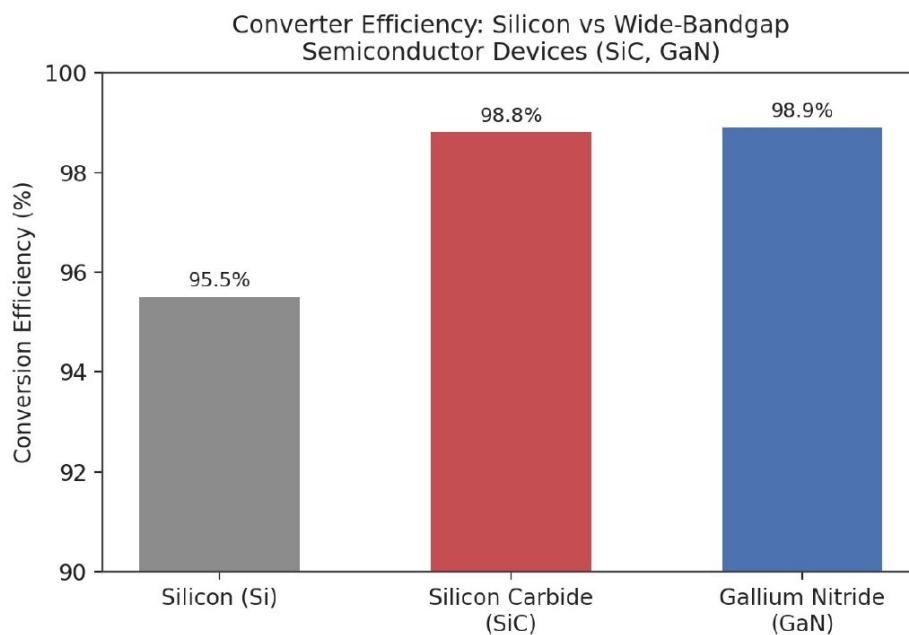


Figure 2. Conversion-efficiency comparison of silicon, silicon-carbide (SiC), and gallium-nitride (GaN) power converters.

Bangladesh National Energy Profile

Global electricity demand has grown steadily over the past three decades, and the composition of global generation is shifting toward renewable sources, a transition made possible by power-

electronic conversion equipment. The figures below, compiled from IEA data for Bangladesh, illustrate representative national-level trends in electricity consumption, emissions, and cross-border trade.



Figure 3. Final consumption of electricity by sector, Bangladesh, 2023 (Source: IEA).

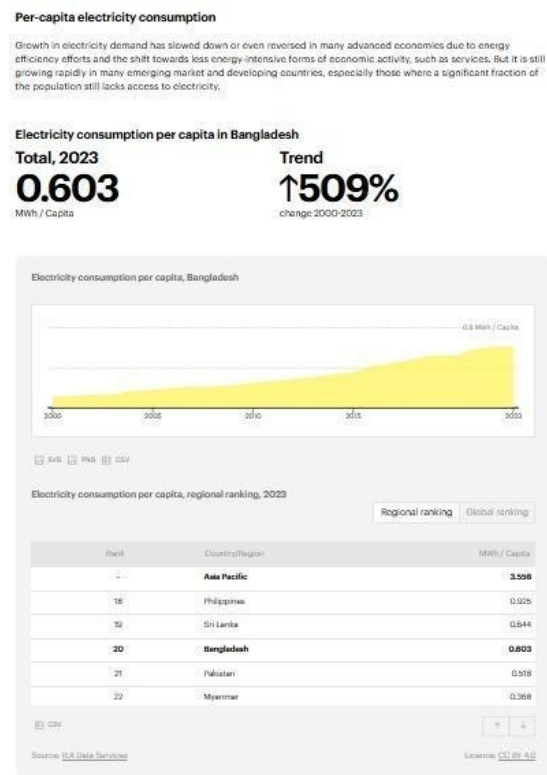


Figure 4. Per-capita electricity consumption and regional ranking, Bangladesh, 2023 (Source: IEA).

CO2 emissions from power generation

Power generation, which includes electricity and heat, is one of the largest sources of CO2 emissions globally, primarily from the burning of fossil fuels like coal and natural gas in thermal power plants.

Share of power in emissions

56%

of total energy-related CO2 emissions, 2023

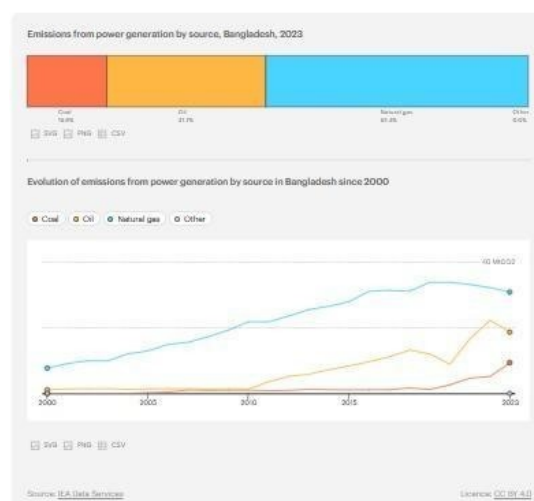


Figure 5. CO2 emissions from power generation by source, Bangladesh, 2023 (Source: IEA).

Electricity imports and exports

Unlike other energy commodities such as coal, oil and natural gas, electricity trade between countries is relatively limited as it is more technically complex and requires a direct cross-border interconnection. Such connections can help to balance out supply and demand across regions, which will be increasingly important as variable renewables like solar and wind make up a larger share of electricity generation.

Net electricity imports

10.7%

of 2023 total electricity final consumption

Trend in electricity imports

↑360%

change 2014-2023

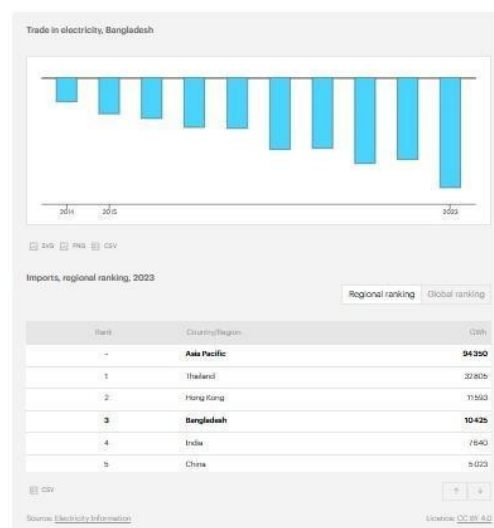


Figure 6. Electricity imports and their trend, Bangladesh, 2014-2023 (Source: IEA).

Electricity is primarily used for heating, cooling, lighting, and to power devices, appliances and industrial equipment; in Bangladesh electricity accounted for 25 percent of total final energy consumption in 2023, with the residential (45%) and industrial (41%) sectors as the largest consuming categories. Electricity consumption per capita reached 0.603 MWh in 2023, a rise of 509 percent since 2000, though this remains low by regional standards. Power generation contributes 56 percent of Bangladesh's total energy-related CO2 emissions, which stood at 106 Mt CO2 (0.31% of global emissions) in 2023, up 429 percent since 2000. Net electricity

imports reached 10.7 percent of final consumption, a rise of 360 percent since 2014, reflecting Bangladesh's growing reliance on cross-border HVDC interconnections to help cover shortfalls.

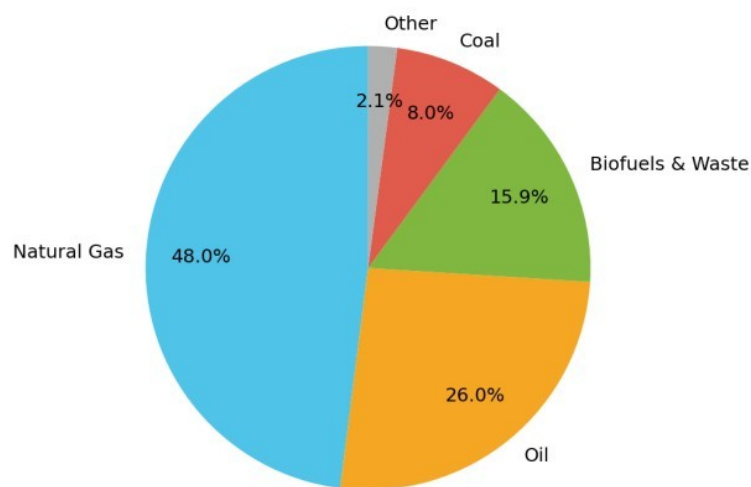


Figure 7. Bangladesh total energy supply mix, 2023 - natural gas 48.0%, oil 26.0%, biofuels and waste 15.9%, coal 8.0%, other 2.1% (Source: IEA).

Bangladesh has made substantial progress in electricity access, with coverage rising from about 20 percent of the population in 2000 to roughly 85 percent today, but the generation mix remains dominated by fossil fuels, principally natural gas, which alone supplies 48 percent of total energy.

Renewable Energy Capacity and Generation Mix

Within the electricity mix, renewable sources still account for a small share of total generation - about 1.9 percent of power generation and only around 3.6 percent when measured against grid electricity supplied - though solar photovoltaic now dominates the renewable sub-mix, supplying 66 percent of total renewable generation.

Solar PV
66%
 of total generation from renewables

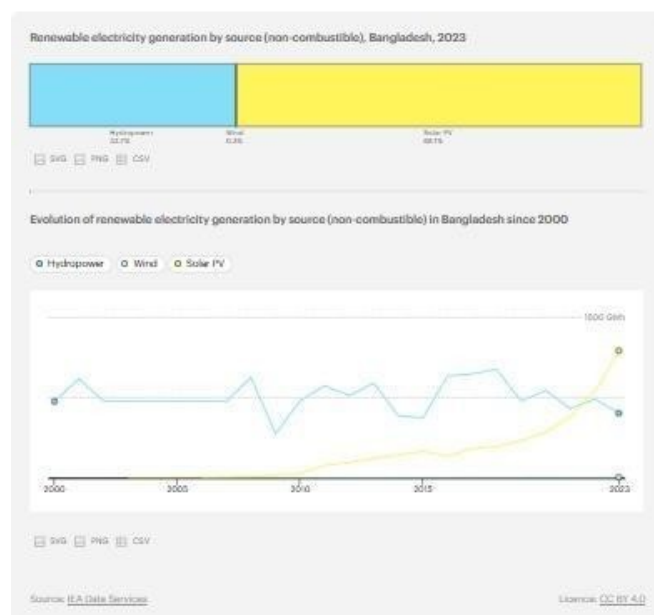


Figure 8. Solar PV share of renewable electricity generation, Bangladesh (Source: IEA).

At the national level, BPDB/SREDA reported a total renewable-energy installed capacity of 1,823.57 MW, of which solar contributes 1,530.48 MW (83.9%), hydropower 230 MW (12.6%), wind 62 MW, and biogas/biomass a marginal share, as summarised in Figure 9.

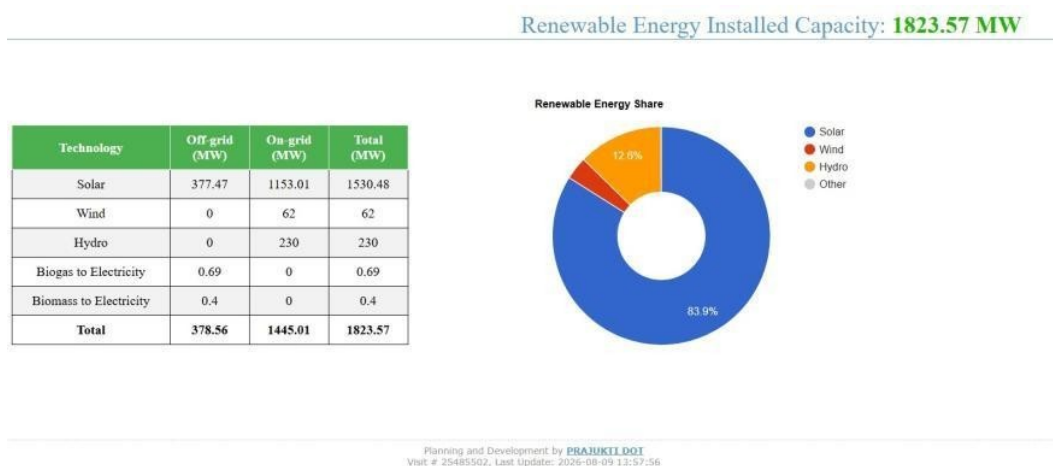


Figure 9. Renewable-energy installed capacity by technology and off-grid/on-grid split, Bangladesh (Source: BPDB/SREDA).

National Grid Generation Capacity

The Bangladesh Power Development Board reported a total installed generation capacity of 29,593 MW as of August 2026, dominated by gas (42.15%), coal (28.46%), and heavy fuel oil (19.06%), as shown in Figure 10.

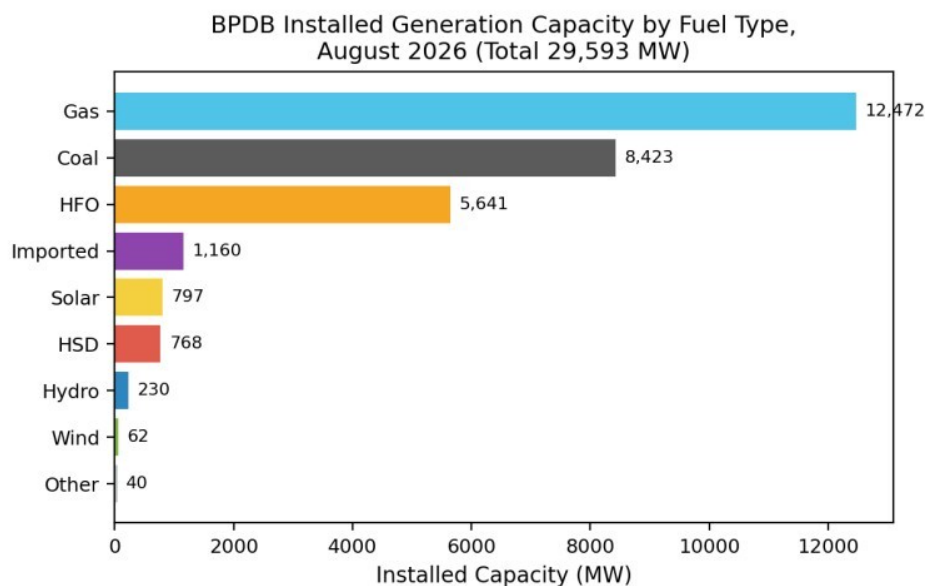


Figure 10. Installed capacity of BPDB power plants by fuel type, August 2026 (Source: BPDB).

Installed capacity, however, overstates what the system can actually deliver: ageing units, water and fuel limitations, and technical constraints reduce the derated (dependable) capacity to 28,793 MW, a difference of roughly 800 MW concentrated mainly in the gas and coal fleets, as shown in Figure 11.

Derated Capacity of BPDB Power Plants as on August 2026

Fuel Type	Capacity(Unit)	Total(%)
Gas	12154 MW	42.21 %
HFO	5637 MW	19.58 %
HSD	768 MW	2.67 %
Coal	7945 MW	27.59 %
Imported	1160 MW	4.03 %
Solar	797 MW	2.77 %
Hydro	230 MW	0.80 %
Wind	62 MW	0.22 %
none	40 MW	0.14 %
Total	28793 MW	100.00 %

Figure 11. Derated (dependable) capacity of BPDB power plants by fuel type, August 2026 (Source: BPDB).

Sector-wise, the public sector and independent private producers each supply roughly 40 percent of total installed capacity, with the remainder coming from joint ventures and cross-border power imports, as shown in Figure 12.

ally, Figure 10 shows the sector-wise present installed generation capacity of Bangladesh. The public sector and independent private producers each supply roughly 40 percent of total capacity, with the remainder from joint ventures and cross-border power imports.

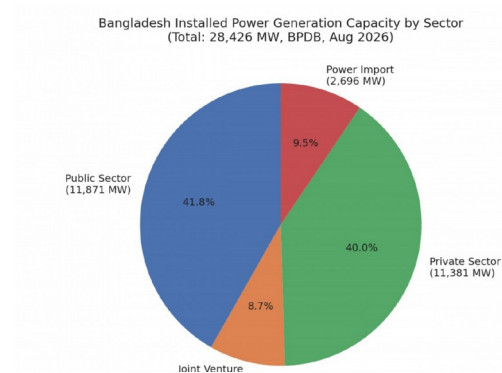


Figure 12. Bangladesh installed power-generation capacity by ownership sector, August 2026 (Source: BPDB).

Peak Demand, Supply, and Load-Shedding

Despite this installed and derated capacity, actual peak generation is frequently constrained by fuel shortages and maintenance outages, producing a persistent gap between demand and supply. On 12 August 2026, the national evening-peak demand reached 18,106 MW against a delivered supply of only 15,568 MW, leaving a load-shed of 2,538 MW distributed unevenly across the country's nine power-distribution zones, with Dhaka, Mymensingh, and Rajshahi bearing the largest shortfalls.

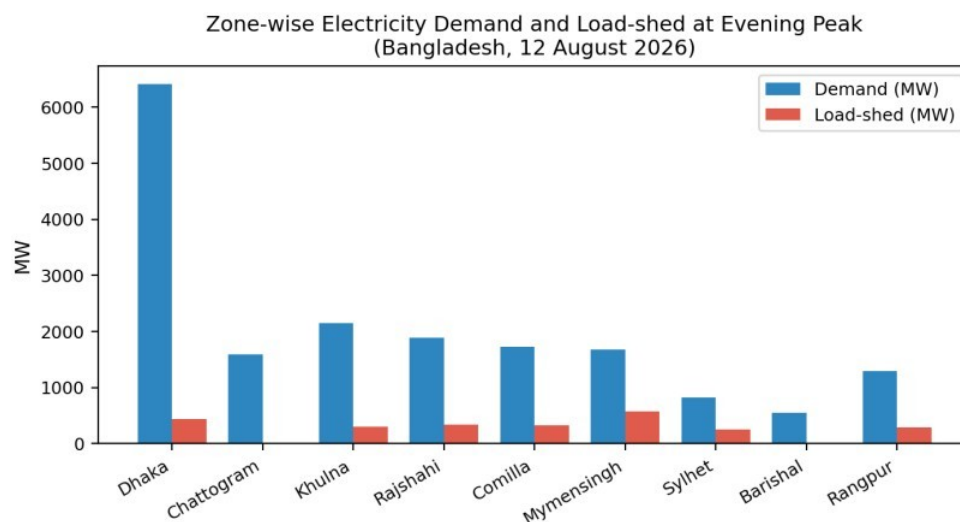


Figure 13. Zone-wise electricity demand and load-shed at evening peak, Bangladesh, 12 August 2026 (Source: PGCB).

A similar pattern is visible in area-wise peak-generation records: the Dhaka region alone accounts for roughly 37 percent of total peak supply, as shown in Figure 14.

13,568	08-08-2026	13:00	4,980.0	1,390.0	1,120.0	1,510.0	480.0	360.0	890.0	1,350.0	1,100.0
16,540	07-08-2026	21:00	6,120.0	1,620.0	1,350.0	1,810.0	590.0	430.0	1,080.0	1,680.0	1,280.0
13,170	07-08-2026	13:00	4,850.0	1,320.0	1,080.0	1,460.0	460.0	350.0	860.0	1,300.0	1,050.0
16,210	06-08-2026	21:00	6,010.0	1,590.0	1,310.0	1,780.0	575.0	420.0	1,050.0	1,640.0	1,250.0

Area-wise power supply at peak generation, Bangladesh, 08 August 2026 (F

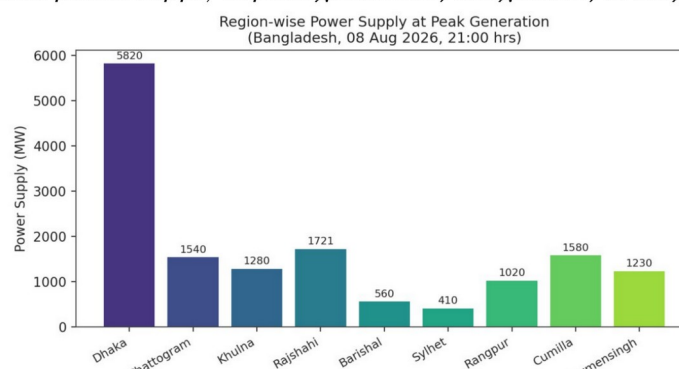


Figure 14. Region-wise power supply at peak generation, Bangladesh, 08 August 2026 (Source: PGCB).

Hourly source-wise records published by PGCB break generation down into gas, liquid fuel, coal, hydro, solar, and wind output, together with cross-border imports from India and Nepal. On 08 August 2026, natural gas and coal remained the dominant sources throughout the day, while solar output followed a clear daylight curve, rising to a midday maximum of about 620 MW and falling to zero after sunset - a pattern that shows directly why inverters and MPPT controllers are indispensable for absorbing the variability of solar and wind generation without destabilising the grid.

07-08-2026	13:00:00	13,170	14,250	1,080	3,537	1,795	4,686	222	600	14	901	154	1,215	38
07-08-2026	12:00:00	12,500	13,550	1,050	3,465	1,223	4,701	222	574	14	901	148	1,206	38

Figure 5. Hourly generation and fuel-mix data, PGCB, 06–08 August 2026 (PGCB).

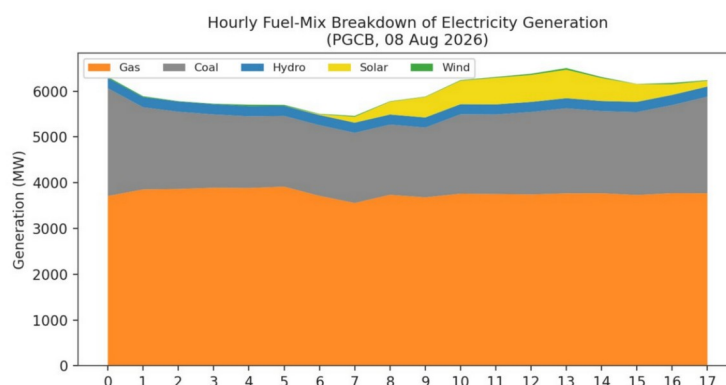


Figure 15. Hourly fuel-mix breakdown of electricity generation, Bangladesh, 08 August 2026 (Source: PGCB).

Demand itself rises from about 14,000 MW overnight to a daytime peak near 15,150-15,200 MW around noon, while available generation lags behind demand throughout the day, producing load-shedding of up to nearly 2,000 MW during the late afternoon even at a supply level exceeding 13,000 MW, illustrating the continuing strain on Bangladesh's generation and transmission infrastructure.

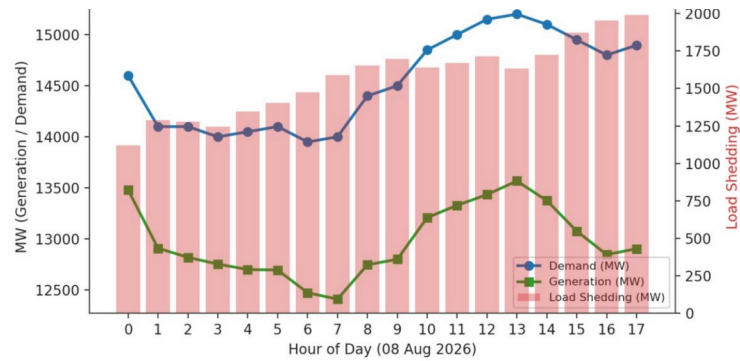


Figure 16. Hourly generation, demand, and load shedding, Bangladesh, 08 August 2026 (Source: PGCB).

Figure 16. Hourly generation, demand, and load-shedding, Bangladesh, 08 August 2026 (Source: PGCB).

Figure 17 shows the day-to-day maximum generation recorded over a thirteen-day window in early August 2026, which fluctuated between about 14,500 MW and just over 16,000 MW as fuel availability and temperature varied.

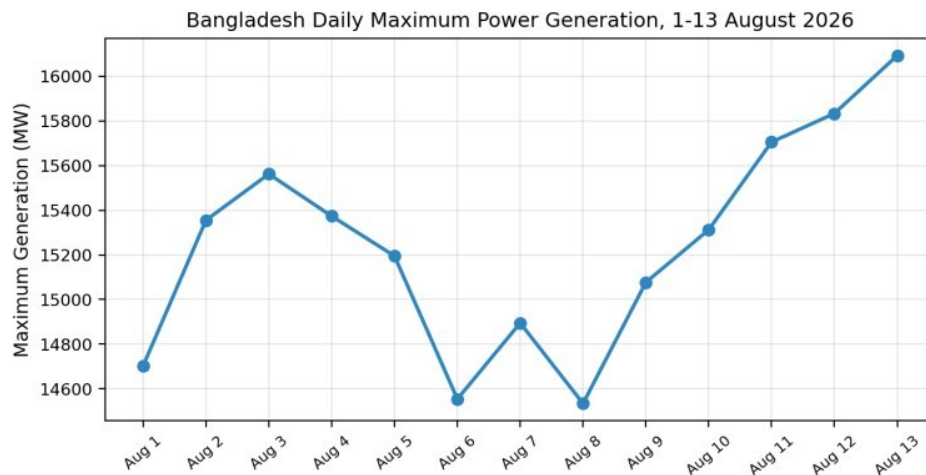
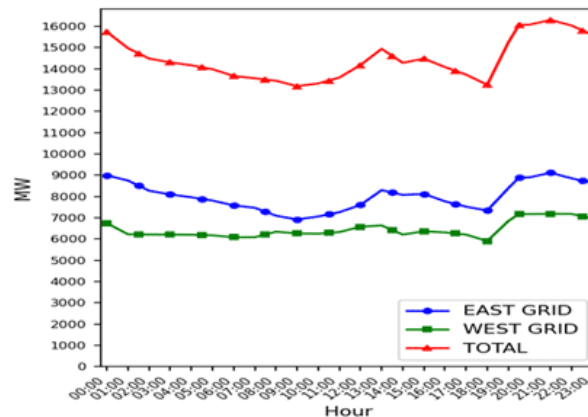


Figure 17. Daily maximum power generation, Bangladesh, 1-13 August 2026 (Source: PGCB).

Figure 18 shows the corresponding daily generation curve recorded by PGCB's National Load Dispatch Centre, in which total generation tracks the sum of the East grid and West grid, peaking in the evening hours as residential and commercial demand rises.

14-08-2026



Executive Engineer
Network Operation Division

Figure 18. Daily generation curve, East grid vs West grid vs total, 14 August 2026 (Source: PGCB, National Load Dispatch Centre).

These figures illustrate why power electronics is particularly important for Bangladesh. Efficient AC-DC-AC conversion in HVDC interconnectors (such as the Bheramara link with India) allows cross-border power imports to help cover shortfalls - though the Adani Power link consistently supplies well below its 1,398 MW scheduled forecast, while the Bheramara HVDC and Nepal links operate closer to their scheduled values. Power-electronic inverters are essential to the growing installed solar capacity already feeding the grid, and modern variable-frequency drives in industrial and irrigation loads can materially reduce the demand that must be met from scarce gas-fired capacity.

Future Prospect

The role of power electronics in the energy sector is expected to expand considerably over the coming decade, closely linked to the global transition toward clean energy and sustainable development:

- Wider adoption of wide-bandgap devices: SiC and GaN devices are expected to see significantly increased use as manufacturing costs fall, improving efficiency in solar inverters, electric vehicles, industrial drives, and data centres.
- Next-generation semiconductor materials: Ultra-wide-bandgap materials such as gallium oxide (Ga₂O₃) and diamond-based semiconductors are being investigated for future high-voltage, high-temperature applications.
- Grid-forming inverters and digitalisation: Grid-forming inverters, solid-state transformers, artificial intelligence, and predictive maintenance are emerging as key enablers for power systems with very high shares of renewable generation.

- Electric-vehicle charging infrastructure: Advanced SiC- and GaN-based converters are expected to improve charging speed, reduce energy losses, and enhance the performance of fast-charging stations.
- Battery storage and vehicle-to-grid systems: Wider deployment of grid-scale storage converters and vehicle-to-grid systems is expected to further support the integration of variable renewable generation.
- Green hydrogen and Power-to-X technologies: High-efficiency electrolyzers that convert renewable electricity into green hydrogen will rely on power-electronic converters to help decarbonise sectors where direct electrification is difficult, alongside continued growth of DC microgrids for rural electrification.

Bangladesh Perspective

Bangladesh's power sector remains heavily dependent on imported natural gas, furnace oil, and coal. Power electronics has nonetheless already enabled significant progress through the Solar Home System (SHS) programme, one of the world's largest off-grid solar initiatives, which uses charge controllers, DC-DC converters, and small inverters to electrify millions of rural households without access to the national grid.

The draft Renewable Energy Policy 2025 targets 20 percent renewable electricity by 2030 and 30 percent by 2041, requiring an estimated investment of USD 35-42.6 billion between 2025 and 2040. Key challenges include limited grid infrastructure and land-acquisition delays, heavy import-dependence for inverters and converters, outstanding payments exceeding USD 2.2 billion to independent power producers, and a shortage of skilled power-electronics engineers. Strengthening local manufacturing, MPPT-based solar systems, grid-connected converters, and technical education will be critical to meeting these targets while improving grid reliability and energy security. As Bangladesh's renewable-energy targets and planned solar parks come online, and as the national grid modernises toward smart-grid operation, power-electronic converters will play an increasingly central role in closing the demand-supply gap sustainably rather than through further fossil-fuel capacity additions.

Conclusion

Power electronics underpins nearly every pathway toward a more sustainable energy system, from grid-scale renewable integration and smart-grid control to electric transport, industrial efficiency, and energy storage. Wide-bandgap devices such as SiC and GaN now achieve conversion efficiencies exceeding 98 percent, and the global market for power electronics in renewable energy continues to expand steadily.

Bangladesh-specific data analysed in this report show that demand continues to rise, fossil fuels still dominate generation, and renewable energy - while growing - remains a modest share of the total mix at under 4 percent of grid electricity. Where natural gas supplies almost half of total energy and evening-peak load-shedding still reaches over 2,500 MW, wider deployment of efficient power-electronic converters, together with continued investment in solar, wind, storage, and grid-interconnection capacity, offers one of the most practical routes to closing the demand-supply gap sustainably. Continued research, policy support, and infrastructure investment - alongside strengthened local technical capability - will be essential for achieving

both energy security and environmental sustainability in Bangladesh and globally in the years ahead.

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